

Ex 8 Conduct Z test for the Data given using Python

```
import numpy as np
from scipy.stats import norm
data = [52, 48, 50, 51, 49, 53, 47, 50, 52, 48]
mu = 50
sigma = 2
n = len(data)
z = (np.mean(data) - mu) / (sigma / np.sqrt(n))
p = 2 * (1 - norm.cdf(abs(z)))
print("Z-value:", round(z, 2))
print("P-value:", round(p, 4))
if p < 0.05:
    print("Reject H0")
else:
    print("Accept H0")
```

```
... Z-value: 0.0
P-value: 1.0
Accept H0
```